TIC 468210255

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period        6.706415 d
  Epoch (t0)    1575.3682 BTJD
  Duration      0.1700 d
  Depth         6.926 %  (69,255 ppm)
  SNR           110.0

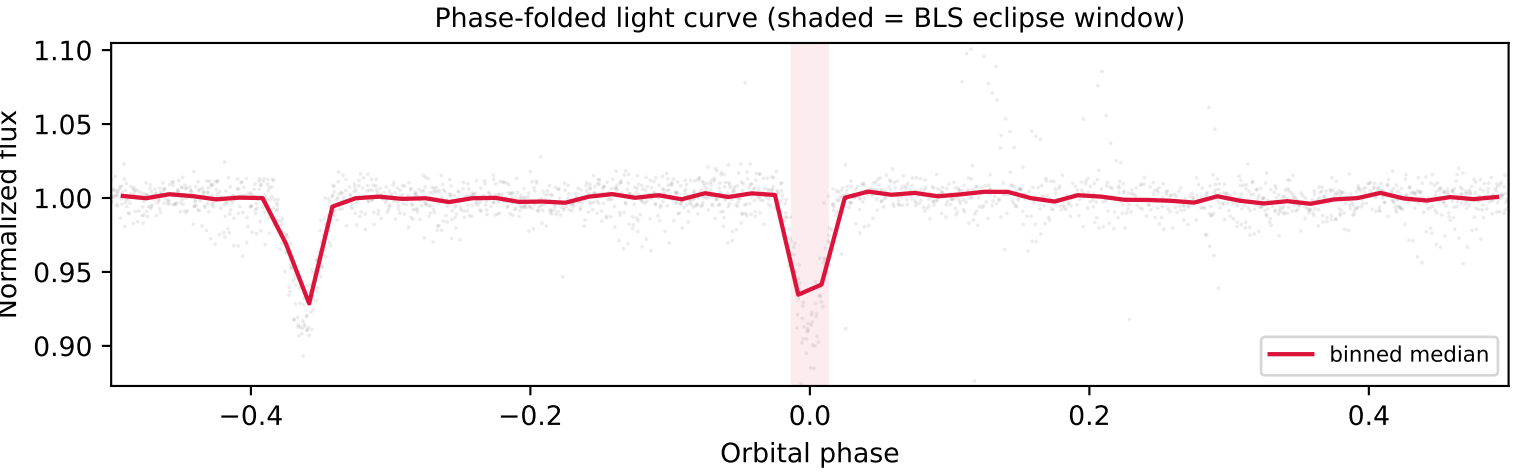
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     7  (gate >= 3)
  Robust depth ratio    1.05  (gate >= 0.50)
  Robust depth sigma    49.9  (gate >= 7.0)
  Odd/even z-score      0.33  (consistent)
  Secondary eclipse     detected, phase -0.36, 7.4 sigma
```

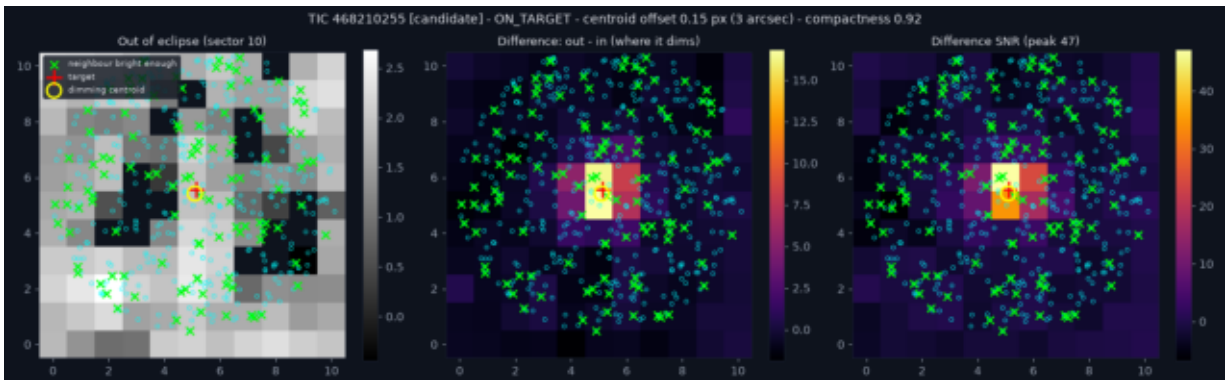
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 10    light-curve dip in this sector: 37.6 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   47.23   (gate >= 5.0)      Compactness   0.92   (gate >= 0.50)
  Dimming centroid offset     0.15 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 468210271
    0.21 px away, dTmag +1.67 - farther from the dimming than the target
```





Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.064 px (dimming centroid 0.149 px from target vs 0.213 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.